Progress Update

Cross-country Dependence and Forecast Value in Daily COVID-19 Case Counts

Refocused around two research questions:
1. identify dependence patterns
2. test whether those patterns add measurable forecast value beyond country-specific benchmarks.

Group 3

Project focus and relevance

Tighter scope

The project now focuses on one connected theme: cross-country dependence and whether it has practical forecast value.

Why COVID-19 still works

COVID-19 is not the end goal of the project. It is used as a rich multi-country empirical setting to study dependence, forecast value, and early-warning ideas that still matter for influenza, RSV, and future outbreaks.

More concrete value

The output is framed for a public-health forecasting lead: should external country signals be monitored as early-warning inputs or treated cautiously?

Feasible capstone

The workload remains substantial through alignment, transformation, dependence modeling, rolling evaluation, and sensitivity checks - without splitting effort across two disconnected questions.

Decision context and value added

Decision context

We are supporting short-horizon infectious-disease monitoring for public-health decision makers who must decide whether signals from other countries are worth tracking in real time.

However, apparent international co-movement can be misleading.
Countries differ in timing, reporting, testing, and wave intensity, so cross-country information should only be used operationally if it shows stable and interpretable value.

Value added to the stakeholder

- If stable lead-lag structure exists, some countries may serve as sentinel indicators.
- If dependence adds forecast value, multivariate monitoring is worth the extra complexity.
- If dependence is weak or unstable, country-specific models should remain the default.
- Either way, the result informs how much weight to place on external epidemic signals.

Revised research questions and narrative flow

RQ1

What cross-country dependence patterns can be identified in aligned daily COVID-19 case-count series?

Goal: identify whether any cross-country signal is stable, directional, and potentially useful for prediction, without making structural causal claims.

RQ2

When and to what extent does cross-country dependence provide forecast value beyond country-specific benchmark models?

Goal: treat forecasting as the operational test of whether dependence is useful, not as a separate disconnected branch.

Dependence patterns -> Forecast value -> Surveillance recommendation

Why COVID-19 remains a useful empirical setting

Key point

The project is not arguing that COVID-19 is still the main infectious disease of concern.

Instead, COVID-19 provides a uniquely rich multi-country surveillance panel for studying how epidemic signals align, diverge, and potentially inform one another across locations.

Why the lessons still generalize

- It offers long, high-frequency, publicly reported case-count series across multiple countries.
- It contains repeated waves, heterogeneous timing, and nontrivial reporting artifacts - exactly the kinds of features that challenge operational surveillance.
- Methods for judging when external regional signals are useful can transfer to settings such as influenza, RSV, or future emerging outbreaks.
- This makes COVID-19 a methodological test bed rather than the final public-health target.

Data scope and preprocessing design

Scope

- **Seven country-level daily case-count series**: Australia, Brazil, Canada, China, South Africa, the United Kingdom, and the United States.
- We analyze both **raw daily case counts** and **cases per 100,000 population**. The per-capita measure improves cross-country comparability by adjusting for large differences in population size.
- **Main variables** are time, location, and case counts; the sophistication comes from temporal alignment, scale standardization, and multivariate structure, rather than from many covariates.
- All analysis is run on an aligned common panel after cleaning, transformation, and standardization.

Preprocessing pipeline

1. **Filter and parse** — Retain the selected source, parse dates and counts, and drop invalid records.
2. **Standardize reporting intervals** — Convert cumulative or interval reports into a reproducible daily series.
3. **Align countries** — Keep the common observation window across countries to support comparable dependence analysis.
4. **Transform for modeling** — Use log(1 + cases) and differencing / standardization as needed for stability and interpretation.

RQ1 analysis plan: identifying dependence patterns

1

Describe cross-country dependence

Plot standardized series, pairwise correlations, rolling correlations, and cross-correlations across lags.

2

Test lead-lag timing

Check whether changes in some countries systematically precede changes in others rather than assuming synchrony.

3

Model directional linkage

Fit a parsimonious VAR only after transformation, lag selection, and stability checks; use Granger tests for predictive linkage.

4

Interpret cautiously

Treat any linkage as predictive content in lagged history, not as structural causality; only show IRFs if the system is stable and interpretable.

RQ2 analysis plan: when dependence adds forecast value

Benchmark side

- Country-specific benchmarks: ETS and ARIMA after the same transformations used in the dependence analysis.
- Rolling or expanding windows preserve time order.
- Primary horizon: 7 days ahead, evaluated with RMSE and MAE.

Forecast-value test

Cross-country dependence only matters operationally if it improves forecasting beyond those benchmarks.

Main candidate: parsimonious VAR(p) on the aligned transformed panel.

Optional secondary check: a **dynamic factor model** only if strong common structure appears and dimensionality needs to be reduced.

Decision rule

- Keep the multivariate model only if gains are systematic across rolling windows, not just isolated wins.
- Report when gains appear: which countries, which phases, and whether value concentrates at short horizons.
- If gains are weak or unstable, conclude that cross-country dependence is cross-country dependence is analytically interesting but not operationally useful for forecasting for forecasting.

Feasibility, risks, and safeguards

Why this is capstone-level work

- **Multi-country data harmonization**, including date alignment, reporting normalization, and handling inconsistent reporting conventions.
- **Repeated modeling choices** on transformations, stationarity, and lag structure for a jointly analyzed panel.
- **Layered dependence analysis**, moving from descriptive evidence to multivariate time-series models and predictive linkage tests.
- **Rolling out-of-sample evaluation** against country-specific benchmarks to assess whether multivariate structure adds real forecasting value.
- **Sensitivity checks and stakeholder translation**, so findings remain credible under alternative specifications and are interpretable for decision-making.

Main risks and mitigation

Nonstationarity / regime changes	Use transformations, differencing choices, and sensitivity checks rather than one rigid specification.
Reporting artifacts	Cross-check descriptive patterns against model-based conclusions and avoid overinterpreting noisy lags.
Overfitting in multivariate models	Keep the multivariate specification parsimonious and compare it against simpler country-specific benchmarks.
Weak forecast gains	Be willing to conclude that dependence is interesting structurally but not useful operationally.

Expected deliverables and bottom line

Expected outputs

- Dependence figures showing co-movement, cross-correlation, and lead-lag patterns.
- VAR-based evidence on directional predictive linkage, with IRFs only if stable.
- Forecast-comparison tables for benchmark vs multivariate models.
- A clear recommendation on when cross-country information is worth using in surveillance and short-term forecasting.

Bottom line

The revised project asks one coherent question in two stages:

First, identify whether meaningful cross-country dependence exists. Then, test whether that dependence actually improves forecasts beyond country-specific baselines.